Suggested Project Topics:

1. Polynomial Regression
2. Generalized Linear Models (GLM)
3. Logistic Regression
4. Multinomial Logistic Regression
5. Ordinal Logistic Regression
6. Poisson Regression
7. Ridge Regression
8. Lasso Regression
9. Elastic Net Regression
10. Principal Components Regression (PCR)
11. Partial Least Squares Regression (PLSR)
12. Robust Regression
13. Quantile Regression
14. Spline Regression
15. Generalized Additive Models (GAM)
16. Bayesian Linear Regression
17. Support Vector Regression (SVR)
18. Random Forest Regression
19. k-Nearest Neighbors Regression (k-NN Regression)
20. Neural Network Regression
21. Gaussian Process Regression
22. Negative Binomial Regression
23. Bayesian Lasso Regression
24. Cox Regression
25. Regression with censored data (general)
26. Semiparametric Regression
27. Non-parametric Regression
28. Kernel Regression
29. Artificial Neural Networks
30. Classification and Regression Trees

1. **Explain the method(s) theoretically**, including:
    - model formulation,
    - assumptions,
    - estimation procedure,
    - strengths and limitations.
2. **Review academic sources**:
    - at least **one book (if possible)**,
    - at least **five peer-reviewed paper**.
3. **Apply the method to a real/syntetic dataset**:
    - justify dataset choice,
    - perform data preprocessing,
    - implement the model (R or Python).

4. **Evaluate model performance** using appropriate metrics.
5. **Interpret results** and discuss practical/methodological implications.

Technical Requirements

- Software: **R or Python**
- All sources must be **properly cited**

Expected Deliverables

- **Written report** (PDF, ~20–50 pages in total)
- **Code file or notebook**
- **Presentation** (30 min.)

Assessment Criteria

Component	**Weight**
Theoretical understanding	30%
Correct implementation	30%
Interpretation & discussion	25%
Clarity & structure	15%